



SpiderRock V7 Tool Concept Guide

Control Viewer

December 3, 2020

OVERVIEW

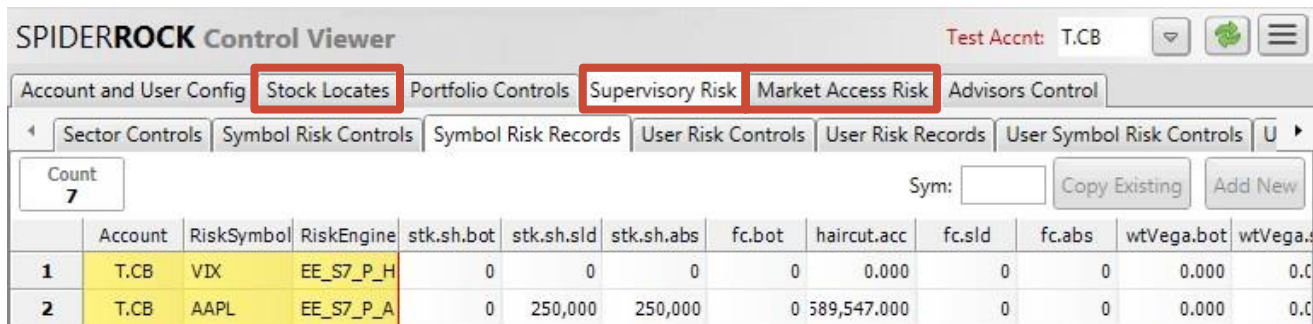
Clients with special authorization can implement, edit, and manage supervisory risk limits and compliance requirements (locates, restricted lists etc) in real time via the Control Viewer as well as our SRSE database (SRControl). This document will provide screenshots of the GUI tool and corresponding SRSE queries.

USER TYPES

SpiderRock enables various types of users to help client firms control who has oversight over which process when using the SpiderRock system. The user categorization is as follows:

ClientTrader:	Can access GUI Tools and SRSE to view/enter/cxl orders in assigned accounts; Cannot edit any risk settings
ClientRiskTrader:	Can access GUI Tools and SRSE to view/enter/cxl orders in assigned accounts; Can edit account risk settings
ClientRiskAdmin:	Can access GUI Tools and SRSE to view/cxl orders in assigned accounts; Can edit account risk and MAR limits for assigned riskFirm; Cannot enter orders
ClientViewOnly	Can access GUI Tools and SRSE to view orders in assigned production or test accounts
ClientSRSEOnly	Can only connect to SRSE but cannot enter any orders
ClientTestOnly	Can access GUI Tools and SRSE for test order entry purpose only
CIStageReviewTrader	Can create an order via the GUI tools that is sent to a client's system; if the client's system approves the order then the order is immediately activated
CIStageReleaseTrader	Can only release staged orders that have been sent to the system from SRSE or FIX
CIStageModifyTrader	Can submit a staged order through SRSE or GUI tools; this order can be modified and released by a stage release trader
CIStageLockTrader	Can submit a staged lock order through SRSE or GUI tools; the order can be modified by a stage release trader, but they cannot increase the size of the order

ACCESSING RISK CONTROLS



Count	Account	RiskSymbol	RiskEngine	stk.sh.bot	stk.sh.sld	stk.sh.abs	fc.bot	haircut.acc	fc.sld	fc.abs	wtVega.bot	wtVega.s
1	T.CB	VIX	EE_S7_P_H	0	0	0	0	0.000	0	0	0.000	0.0
2	T.CB	AAPL	EE_S7_P_A	0	250,000	250,000	0	589,547.000	0	0	0.000	0.0

The Control Viewer is organized in tabs and sub tabs.

Account and User Config: Provides user configuration information (User Config), route information (AltRoute Config), theoretical model descriptions (TheoModel Config).

Stock Locates: Users can upload a list of locates (ticker + quantity) so that the SpiderRock system automatically checks locates in a short sale situation. Refer to page 10 of this concept guide for more information.

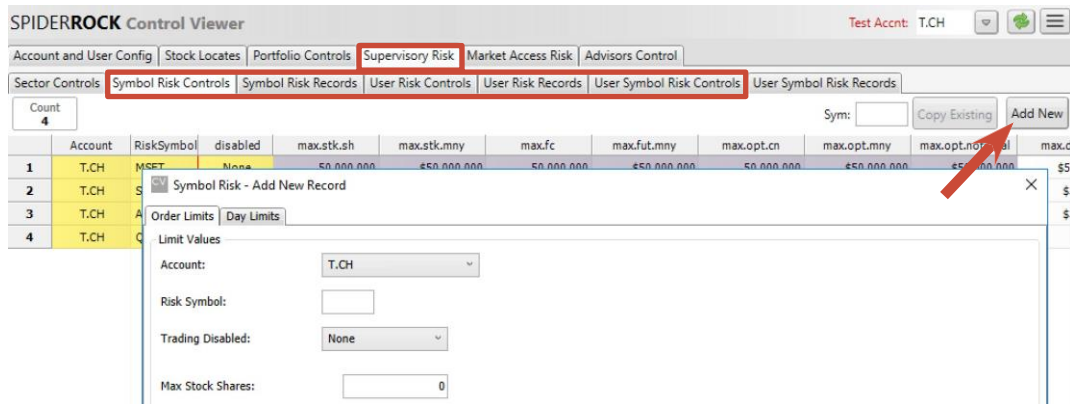
Portfolio Controls (mostly for systematic traders): Enables traders to implement risk limits at the book level, industry level, expiration level as well as risk class (custom sector) level. The symbol control sub tab enables traders to control the trading status at the symbol level (buy only, sell only, etc) as well as the hedge delta rule.

Supervisory Risk: This tab allows the user to set risk controls at the symbol level, user level, or user/symbol level.

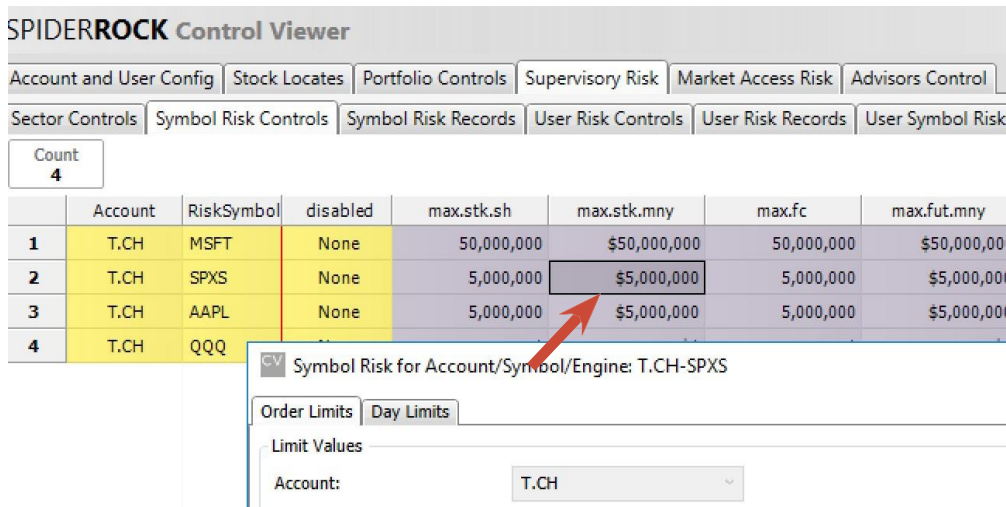
- Controls sub tabs: Clients can implement and edit supervisory risk limits
- Records sub tabs: Clients can view the current consumption of limits during a trading period

Market Access Risk Tab: MAR controls are set at the firm level during the on-boarding process and can be edited directly by authorized users. Authorized risk managers/traders can also submit a request to the SpiderRock support desk to update these limits intraday by sending an email at cs@spiderrock.net.

EDITING SUPERVISORY ORDER RISK LIMITS



- Under the Supervisory Risk tab, in any of the 'Controls' sub tabs users can implement ORDER limits at the Symbol level, User level, or User/Symbol level (using the appropriate sub-tab) by clicking on the “Add New” button (see red arrow above). Users can also edit existing limits by double-clicking on any values in the grid (see red arrow below).



- The order limits are the same for all sub-tabs and are listed below:
 - Max number of shares
 - Max stock money
 - Max number of futures contracts
 - Max futures money
 - Max number of option contracts
 - Max option money (Premium * Contracts * Multiplier)
 - Max option notional (Underlier Price * Contracts * Multiplier)
- Users save entries by clicking on 'Post changes' in the lower right corner of the window

EXAMPLE OF ORDER LEVEL CONTROL IN SRSE

SRControl - MySQL query equivalent

The risk controls at the order level can also be implemented via SRControl, which is one of the SRSE live databases. The SQL query below would insert risk limits at the order level of 10,000 shares, 1 million in stock money, 100 option contracts, and 1 million in option money:

```
INSERT INTO srcontrol.msgsrriskcontrol (acct, riskSymbol, disabled, maxStkShares,
maxStkMoney, maxFutContracts, maxFutMoney, maxOptContracts, maxOptMoney,
maxOptNotional, maxDayHaircut, maxDayDDeltaLn, maxDayDDeltaSh, maxDayDDeltaAbs,
maxDayVegaLn, maxDayVegaSh, maxDayVegaAbs, maxDayWtVegaLn, maxDayWtVegaSh,
maxDayWtVegaAbs, maxDayNotionalLn, maxDayNotionalSh, maxDayNotionalAbs,
maxDayStkShAbs, maxDayFutCnAbs, maxDayOptCnAbs, userName, TIMESTAMP)
```

```
VALUES ('T.acct', 'DFLT', 'No', 10000,1000000,0,0,100,1000000,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,-1,'username',NOW());
```

EDITING SUPERVISORY DAY LIMITS

SPIDERROCK Control Viewer

Test Acct: T.CH

Account and User Config | Stock Locates | Portfolio Controls | Supervisory Risk | **Market Access Risk** | Advisors Control

Sector Controls | **Symbol Risk Controls** | Symbol Risk Records | User Risk Controls | User Risk Records | User Symbol Risk Controls | User Symbol Risk Records

Count: 4

Symbol Risk - Add New Record

Order Limits | Day Limits

Limit Values

Type	RISK LIMITS			CURRENT VALUES		
	Bought	Sold	Absolute	Bought	Sold	Absolute
Dollar Delta:	100,000	0	0	0	0	0
WtVega:	0	0	0	0	0	0
Vega:	0	0	0	0	0	0
Notional:	0	0	0	0	0	0
Haircut:			0			0

Copy Existing | Add New

- Under the Supervisory Risk tab, in any of the 'Controls' sub tabs users can implement DAY limits at the Symbol level, User level, or User/Symbol level (using the appropriate sub-tab) by clicking on the "Add New" button (see red arrow above). Users need to select the 'Day Limits' tab in order to implement DAY limits. Users can also edit existing limits by double-clicking on any values in the grid (see red arrow below).

SPIDERROCK Control Viewer

Test Acct: T.CH

Account and User Config | Stock Locates | Portfolio Controls | Supervisory Risk | Market Access Risk | Advisors Control

Sector Controls | Symbol Risk Controls | Symbol Risk Records | User Risk Controls | User Risk Records | User Symbol Risk Controls | User Symbol Risk Records

Count: 4

Symbol Risk for Account/Symbol/Engine: T.CH-MSFT

	Account	RiskSymbol	disabled	max.stk.sh	max.stk.mny	max.fc	max.fut.mny	max.opt.cn	max.opt.mny	max.opt.notion
1	T.CH	MSFT	None	50,000,000	\$50,000,000	50,000,000	\$50,000,000	50,000,000	\$50,000,000	\$50,000,000
2	T.CH	SPXS	None	5,000,000	\$5,000,000	5,000,000	\$5,000,000	5,000,000	\$5,000,000	\$5,000,000
3	T.C									
4	T.C									

Order Limits | Day Limits

Limit Values

Type	RISK LIMITS			CURRENT VALUES		
	Bought	Sold	Absolute	Bought	Sold	Absolute
Dollar Delta:	50,000,000	50,000,000	50,000,000	0	0	0

- The day limits are listed below:
 - Max Dollar Delta (bought/sold/absolute)
 - Max Time Weighted Vega (bought/sold/absolute)
 - Max Vega (bought/sold/absolute)
 - Max Notional (bought/sold/absolute) as sum of premiums
 - Max Haircut (refers to an equivalent TIMS margin calculation with a \$25 (JBO) or \$37.50 (PM) minimum based on day trades)
 - Max number of shares
 - Max number of futures contracts
 - Max number of option contracts
- Live values can be monitored from the CURRENT VALUES section.

EXAMPLE OF DAY LEVEL CONTROL IN SRSE

SRControl - MySQL query equivalent

The day risk controls can also be implemented via SRControl, which is one of the SRSE live databases. The SQL query below would insert **maximum daily limits** for a particular symbol and user of 100 million dollar delta bought/sold and 200 million absolute dollar delta, a max 50,000 day vega bought/sold and 100,000 absolute day vega, a max of 500 million day notional bought/sold and a 750 million absolute day notional.

```
INSERT INTO srcontrol.msgusersymriskcontrol (acctnt, riskUser, riskSymbol, disabled,
maxStkShares, maxStkMoney, maxFutContracts, maxFutMoney, maxOptContracts, maxOptMoney,
maxOptNotional, maxDayHaircut, maxDayDDeltaLn, maxDayDDeltaSh, maxDayDDeltaAbs,
maxDayVegaLn, maxDayVegaSh, maxDayVegaAbs, maxDayWtVegaLn, maxDayWtVegaSh,
maxDayWtVegaAbs, maxDayNotionalLn, maxDayNotionalSh, maxDayNotionalAbs,
maxDayStkShAbs, maxDayFutCnAbs, maxDayOptCnAbs, userName, TIMESTAMP)
```

```
VALUES ('T.accnt', 'username', 'AAPL', 'No', -1, -1, -1, -1, -1, -1, -1, 1000000000, 1000000000,
2000000000, 50000, 50000, 100000, -1, -1, -1, 5000000000, 5000000000, 750000000,
-1, -1, -1, 'username', NOW());
```


MANAGING MAR RISK LIMITS

The screenshot shows the SPIDERROCK Control Viewer interface. The 'Market Access Risk' tab is selected. A red arrow points to the 'Add New' button in the top right corner of the grid area.

Count	allowMarketOnClose	allowLimitOnClose	allowShortSaleExempt	allowInterMarketSwee	allowOddLots	creditLimitAcc	creditLimitDay	orderLimitStkShares	orderLimitFutContracts
1	None	None	None	None	None	\$0	\$0	0	0
2	None								
3	None								
4	No								
5	None								
6	No								
7	None								
8	None								
9	None								
10	None								

- The Market Access Risk tab is view-only for most users and requires special authorization to make edits. The limits are global and are applied at the account level. Only 'Client Risk Admin' users can edit these limits.
- Under the Market Access Risk tab, authorized users can edit rules and MAR limits by clicking on the "Add New" button (see red arrow above) and selecting the 'Rules' or 'Limits' tab. Authorized users can also edit existing limits by double-clicking on any values in the grid (see red arrow below).

The screenshot shows the 'Limits' tab for a specific risk firm. A red arrow points to the value '\$300,000,000' in the 'CURRENT VALUES' section.

RiskFirm	RiskGroup	RiskEngine	Enabled	Exposure Def	allowMarketOnClose	allowLimitOnClose	allowShortSaleExempt	allowInterMarketSwee	allowOddLots	creditLimitAcc	creditLimitDay	orderLimitStkShares
1										\$0	\$0	0
2										\$0	\$0	0
3										\$0	\$0	0
4										\$0	\$0	0
5										\$0	\$0	0
6										-\$1	\$300,000,000	1,000,000
7										\$0	\$0	0
8										-\$1	\$50,000,000	100,000
9										\$0	\$0	0
10										\$0	\$0	0
11										\$0	\$0	0
12										\$0	\$0	0
13										-\$1	\$25,000,000	0

- The MAR limits are listed below:
 - Max Number of shares/contracts per order
 - Max Share Money per order
 - Max Contract Money per order
 - Max day/book Haircut (refers to an equivalent TIMMS margin calculation with a \$25(JBO) or \$37.50(PM) minimum based on day trades)
 - CreditLimitDay (day trades)
 - CreditLimitAccount (Open Positions + day trades))
 - Max Total Exposure (haircut + open orders exposure)
- Current consumption can be checked under the CURRENT VALUES section.

REQUESTING/UPLOADING LOCATES

An account can be configured to either "Require Locate: Yes/No". If No is selected, SpiderRock bypasses the locate process completely, relying on the executing broker to implement the check on their side. If Yes, SpiderRock will not send a short sale order without a locate.

- **Automatic file loading to our SFTP site or directly into SRSE (srcontrol.msgawaystocklocategateway):** SpiderRock can automate the upload of locate files directly from the client's clearing broker with proper authorization.
- **Control Viewer Stock Locates tab:** Users can request locates if they have an established locate request gateway connection (currently only with ABN) by clicking on the Request Locates button. More commonly, users can manually upload previously authorized locates via the Upload Locates button.



- **Symbol Viewer for intra-day edits:** Similarly, users can also request or manually upload a locate record directly from the Symbol Viewer by double clicking on the LOCATE box (see red arrow below). This will open the away Locate window where users can enter Locate Quantity, Locate Firm and Locate ID.

